Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Machine Learning Base Recommendation System for E-Commerce Platforms

Kirti	Lisha Goel	Simran
2210991805	2210990542	2210990850

Supervised By

Dr. Anshu Singla

Department of Computer Science and Engineering,

Chitkara University, Punjab

Introduction

AI and Machine Learning are transforming modern e-commerce through intelligent recommendation systems.

Artificial Intelligence (AI) and Machine Learning (ML) are rapidly changing online shopping experiences by providing personalised product recommendations.

Traditional recommendation systems often fail to understand changing customer preferences and behaviour.

AI-based recommendation systems analyse customer browsing history, ratings, and purchasing behaviour to suggest relevant products.

These systems improve customer satisfaction, engagement, and sales performance.

Problem Statement

Traditional recommendation systems face several limitations in personalised shopping.

- Traditional systems suffer from **cold-start and data sparsity problems**
- Generic recommendations reduce personalisation effectiveness
- Difficulty in analysing large-scale customer behaviour data
- Lack of adaptability to changing user preferences

These challenges create the need for an intelligent recommendation system powered by AI and ML.

Objectives

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This study evaluates the impact of AI-driven recommendation systems in e-commerce.

- To analyse ML and DL-based recommendation systems
- To compare collaborative, content-based, and hybrid recommendation models
- To evaluate customer engagement and personalisation outcomes
- To identify benefits, limitations, and future implications of AI in e-commerce

Literature Review

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Previous studies highlight the effectiveness of AI in recommendation systems.

Recent studies show that **Machine Learning and Deep Learning significantly improve recommendation accuracy**.

Techniques like **Collaborative Filtering, CNN, LSTM, and Hybrid Models** improve personalisation and recommendation quality.

Research indicates that AI-driven systems increase customer retention, engagement, and purchase probability.

However, many studies lack long-term scalability validation and real-world implementation analysis.

Research Gap

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Existing recommendation systems still face several limitations.

- Limited focus on **real-time adaptive recommendations**
- Cold-start and sparse data issues still exist
- Lack of standardised evaluation framework
- Limited research on ethical concerns and transparency

This study highlights the need for smarter, scalable, and explainable recommendation systems.

AI Tools Consideration

Different AI techniques contribute uniquely to recommendation systems.

● **Collaborative Filtering** – Recommends products based on user behaviour similarities

● **Content-Based Filtering** – Suggests products based on product similarity

● **Hybrid Recommendation System** – Combines multiple recommendation techniques

● **Deep Learning Models (CNN, LSTM, BERT)** – Improve recommendation accuracy and personalisation

Methodology

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The study follows a **review-based and comparative approach**.

● Research papers from **IEEE, Springer, MDPI, ScienceDirect, and ResearchGate** were analysed

● Studies published between **2021–2025** were selected

● Comparison between **traditional and AI-based recommendation systems** was performed

● Real-world industrial applications were studied

The aim was to evaluate the effectiveness of AI-driven recommendation systems in e-commerce.

Evaluation Parameters

Both quantitative and qualitative metrics were used.

Quantitative

- Accuracy
- Precision
- Recall
- Recommendation Relevance

Qualitative

- Customer Satisfaction
- Personalisation
- User Experience
- Customer Engagement

A balanced analysis was used to evaluate recommendation performance.

Result

AI recommendation systems show measurable improvement over traditional systems.

● Improved recommendation accuracy (**above 90% in several studies**)

● Better personalisation and relevant product suggestions

● Increased customer engagement and retention

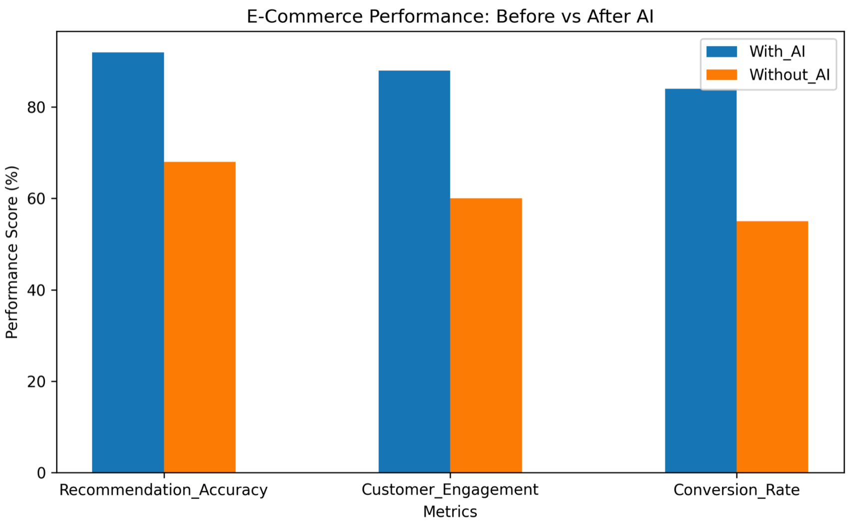
● Improved conversion rates and customer satisfaction

These findings demonstrate the effectiveness of ML and DL in recommendation systems.

Impact of AI on Customer Experience

AI improves customer shopping experiences significantly.

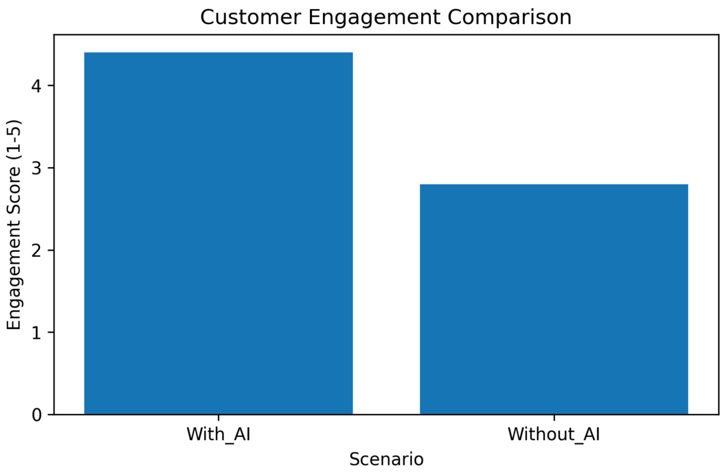
- Personalised shopping recommendations
- Faster and better product discovery
- More relevant suggestions based on preferences
- Improved shopping satisfaction and convenience
- Better understanding of customer needs



Customer Engagement

AI increases customer interaction with e-commerce platforms.

- Higher engagement through personalized recommendations
- Better click-through rates
- Increased customer retention
- Encourages repeat purchases
- Enhances customer loyalty



Impact on Busines Efficiency

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AI improves operational efficiency for e-commerce businesses.

- Automates recommendation processes
- Helps analyze customer behaviour patterns
- Supports data-driven decision making
- Improves scalability of recommendation systems
- Increases business productivity
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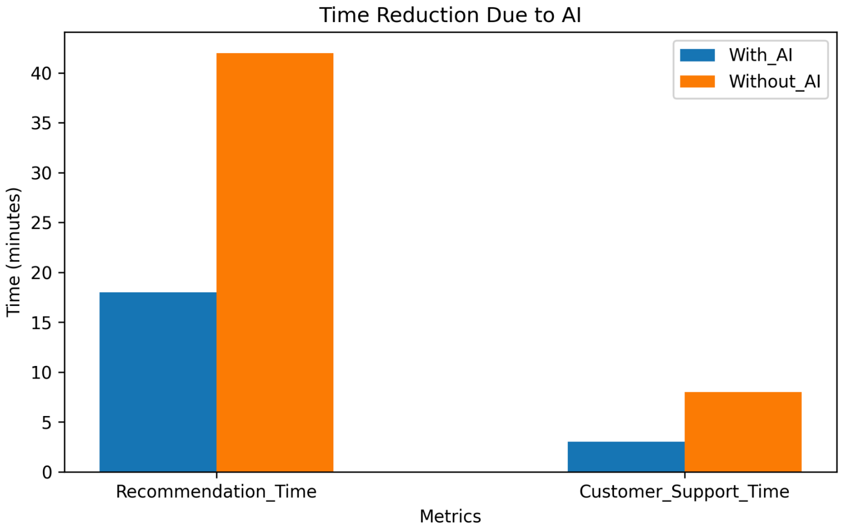


AI Impact on E-Commerce Platforms

AI enables businesses to:

- Improve recommendation quality
- Enhance customer engagement
- Increase sales and conversion rates
- Offer personalised marketing strategies

Major platforms like **Amazon, Walmart, Alibaba, and Netflix** successfully use AI recommendation systems.

Key Advantages

- High recommendation accuracy
- Improved personalisation
- Better customer satisfaction
- Scalability for large customer bases
- Real-time recommendation capability

Challenges & Limitations

Despite benefits, AI recommendation systems face challenges.

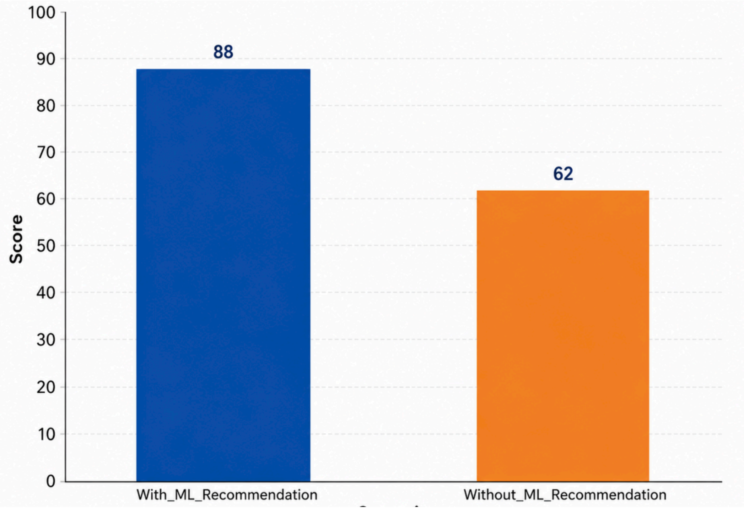
- Data privacy and security concerns
- Algorithm bias and fairness issues
- Cold-start and sparse data problems
- High computational requirements

These challenges must be addressed for responsible AI implementation.

Overall Impact of AI in E-Commerce

AI creates an intelligent and customer-centric shopping ecosystem.

- Improves recommendation quality and personalisation
- Enhances customer engagement and retention
- Supports business growth and decision-making
- Best results achieved with **AI + Human intelligence collaboration**



Conclusion

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AI and Machine Learning significantly improve recommendation systems in e-commerce.

- ML and DL models provide personalised, scalable, and intelligent recommendations.
- Recommendation systems improve customer satisfaction and business performance.
- Ethical and transparent AI implementation is essential.

AI is not replacing human decisions — it is enhancing customer experiences.

Future Scope

Advanced NLP and conversational AI systems

● Integration with **AR/VR and IoT technologies**

● Hyper -personalised recommendation systems

● Improved visual recognition for product recommendations

● Stronger ethical and transparent AI frameworks

Thank You

We welcome your questions and feedback